Approximation of stochastic insurer balance-sheet results using signatures of economic scenarios.

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Quick review

For regulatory purposes, insurers need to value

1. their assets $\rightsquigarrow$ "easy", usually model-independent (existence of liquid markets),
2. their liabilities (BE, VIF) $\rightsquigarrow$ "hard", more challenging task (no more liquid markets).

Following the risk-neutral valuation principle:

$$\text{Fair value of liabilities} = \mathbb{E}[\text{discounted cashflows}] \rightsquigarrow \text{no closed form, requires Monte Carlo approach.}$$

How to sample the discounted cash flows?

$$\text{discounted cashflows} = \Phi(\text{economic scenarios}),$$

where the economic scenarios are sampled from an ESG, and the deterministic mapping

$$\Phi : \text{economic scenarios} \mapsto \text{discounted cashflows},$$

is the so-called ALM model.

The computational bottleneck

Main difficulties:

- ALM models involve complex management rules, profit-sharing mechanisms, surrender behaviors and regulatory constraints $\rightsquigarrow$ strong nonlinearities, barriers, and path-dependence.
- The inherently sequential nature of the computations $\rightsquigarrow$ precludes parallelization across time.
- The presence of conditional mechanisms (e.g. management rules) $\rightsquigarrow$ limits vectorization and efficient implementation.

$$\text{total cost} = \text{cost per scenario} \times \underbrace{\text{number of economic scenarios}}_{\text{large and kept fixed}}$$

reduce the total cost $\rightsquigarrow$ reduce the cost per economic scenario $\rightsquigarrow$ develop a surrogate for the ALM model functional Φ.

What do we approximate?

Let

- $T > 0$ denote the (finite) horizon of all projections,
- $X := (X_t)_{t \in [0,T]}$ denote the (possibly random) economic scenario,
- $\text{CF} = \Phi(X)$ denote the associated discounted cashflows.

We want to approximate the functional $\Phi : (X_t)_{t \in [0,T]} \mapsto \text{CF} \rightsquigarrow$ find another functional $\tilde{\Phi}$ such that $\Phi \simeq \tilde{\Phi}$.

Approximation of a functional defined on a path space!

For a *smooth* path $X := (X_t^1, \cdots, X_t^d)_{t\in[0,T]}$, and for each word $\boldsymbol{i_1 \cdots i_n}$ with $\boldsymbol{i_k} \in \mathcal{A} := \{\boldsymbol{0}, \cdots, \boldsymbol{d}\}$ and $n \in \mathbb{N}^*$ a positive integer, we define:

$$\widehat{\mathbb{X}}_T^{\boldsymbol{i_1}\cdots\boldsymbol{i_n}} = \int_0^T \int_0^{t_1} \cdots \int_0^{t_{n-1}} \mathrm{d}\widehat{X}_{t_n}^{\boldsymbol{i_1}} \cdots \mathrm{d}\widehat{X}_{t_1}^{\boldsymbol{i_n}},$$

and

$$\widehat{\mathbb{X}}_T^{\boldsymbol{\emptyset}} = 1,$$

where $\boldsymbol{\emptyset}$ denotes the empty word, $\widehat{X}_t^{\boldsymbol{0}} = t$ and $\widehat{X}_t^{\boldsymbol{i}} = X_t^{\boldsymbol{i}}$ for $\boldsymbol{i} \in \{\boldsymbol{1}, \cdots, \boldsymbol{d}\}$.

- The time-augmented signature of X is $\widehat{\mathbb{X}}_T := (\widehat{\mathbb{X}}_T^{\boldsymbol{i_1}\cdots\boldsymbol{i_n}})_{(\boldsymbol{i_1},\cdots,\boldsymbol{i_n})\in\mathcal{A}^n, n\in\mathbb{N}}$.
- For $N \in \mathbb{N}$, the N-step truncated time-augmented signature of X is $\widehat{\mathbb{X}}_T^{\leq N} := (\widehat{\mathbb{X}}_T^{\boldsymbol{i_1}\cdots\boldsymbol{i_n}})_{(\boldsymbol{i_1},\cdots,\boldsymbol{i_n})\in\mathcal{A}^n, n\leq N}$ $\rightsquigarrow \widehat{\mathbb{X}}_T^{\leq N} \in \mathbb{R}^{\frac{(d+1)^{N+1}-1}{d}}$ $\rightsquigarrow$ finite family of features.
- Libraries for efficiently computing signatures include: *iisignature* [4], *signatory*, *esig*.

When X is a continuous semimartingale: we define its time-augmented signature using Stratonovich integration theory.

Example. Let $(X_t)_{t\in[0,T]}$ be a *smooth* real-valued path, so that the alphabet is $\{0,1\}$.

Level	Components
0	$\widehat{\mathbb{X}}_T^{\emptyset} = 1$
1	$\widehat{\mathbb{X}}_T^{\mathbf{0}} = \int_0^T \mathrm{d}t = T, \qquad \widehat{\mathbb{X}}_T^{\mathbf{1}} = \int_0^T \mathrm{d}X_t = X_T - X_0$
2	$\widehat{\mathbb{X}}_T^{\mathbf{00}} = \int_0^T \int_0^t \mathrm{d}s\,\mathrm{d}t = \frac{T^2}{2}, \qquad \widehat{\mathbb{X}}_T^{\mathbf{01}} = \int_0^T \int_0^t \mathrm{d}s\,\mathrm{d}X_t = \int_0^T t\,\mathrm{d}X_t,$ $\widehat{\mathbb{X}}_T^{\mathbf{10}} = \int_0^T \int_0^t \mathrm{d}X_s\,\mathrm{d}t = \int_0^T (X_t - X_0)\,\mathrm{d}t, \qquad \widehat{\mathbb{X}}_T^{\mathbf{11}} = \int_0^T \int_0^t \mathrm{d}X_s\,\mathrm{d}X_t = \frac{(X_T - X_0)^2}{2}$

Think of the signature as a polynomial features, but for paths.

Signature of paths: why should this work?

Sufficiently rich linear combinations of signature terms approximate continuous path functionals [2]:

$$\Phi(X) \simeq \underbrace{\langle \beta, \widehat{\mathbb{X}}_T^{\leq N} \rangle}_{\tilde{\Phi}(X)}, \quad \text{for large } N \text{ and some } \beta \in \mathbb{R}^{\frac{(d+1)^{N+1}-1}{d}}$$

nonlinear path-dependent ALM functional $\rightsquigarrow$ linear model in nonlinear path features

$$\beta_N^* := \arg\min_{\beta} \mathbb{E}[|\Phi(X) - \langle \beta, \widehat{\mathbb{X}}_T^{\leq N} \rangle|^2]$$

From ESG to surrogate

The number of features grows like $(d+1)^{N+1} \rightsquigarrow$ problematic when d is large!

Consider an economic scenario (ESG output) containing:

- the risk-free curve across 30 maturities $\rightsquigarrow$ each maturity is encoded by a letter,
- a basket of 5 equities $\rightsquigarrow$ each equity is encoded by a letter,

$$d = 35 \rightsquigarrow \quad 1333 \text{ features } (N=2), \quad 47989 \text{ features } (N=3)$$

Realistic ESG outputs can contain many more coordinates/risk factors (≈ 300)!

$\Downarrow$

A preliminary dimension-reduction step is therefore required

In what follows, $X = (X_t^1, \cdots, X_t^d)_{t\in[0,T]}$ denotes the compressed/reduced economic scenario (d is moderate).

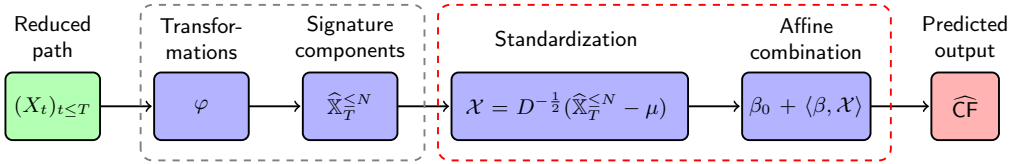


Figure: **Signature-based model.** Parameters: μ, D, β_0, β. Hyper-parameters: φ, N. Blocks circled in red are optimized during training; blocks circled in gray are specified prior to training.

Experimental setup

ALM setup:

1. $\mathsf{CF} = \sum_{t=1}^{T} D(0,t)\,\mathsf{P\&L}_t, \quad \mathsf{VIF} = \mathbb{E}[\mathsf{CF}]$,
2. ALM model: Python implementation of an open source model [1].

ESG setup:

1. 2000 economic scenarios $\rightsquigarrow$ split $70\%/30\%$ into Train/Test,
2. Risk factors:
   - risk-free rate curve $\rightsquigarrow$ 40 maturities,
   - 2 equity indexes,
   - 7 risky rate curves $\rightsquigarrow$ 40 maturities for each rating,
   - rating transition probabilities.

Computing the 2000 cashflow outputs with the full ALM model $\rightsquigarrow$ 3 hours

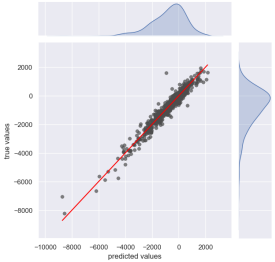
Main results

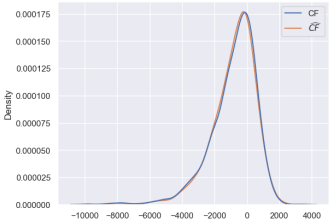
Order	R^2 (%)	Time (s)
2	80.44	0.7
3	92.72	5.8
4	94.63	7.1
5	95.25	56.8

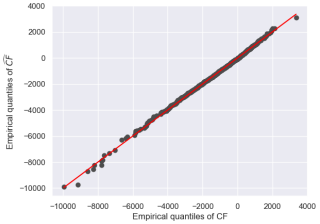
Table: Out-of-sample results for each truncation order.

Surrogate training time (excluding ESG generation)

Main results



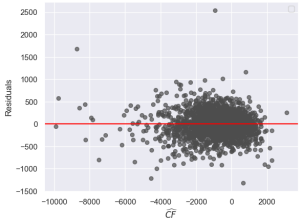


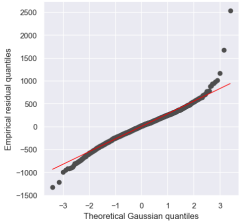


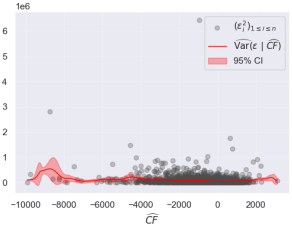
Empirical study of cashflow replication. Results are obtained out-of-sample with the truncation order equal to 4.

VIF relative error $\rightsquigarrow -0.2\%$

Main results

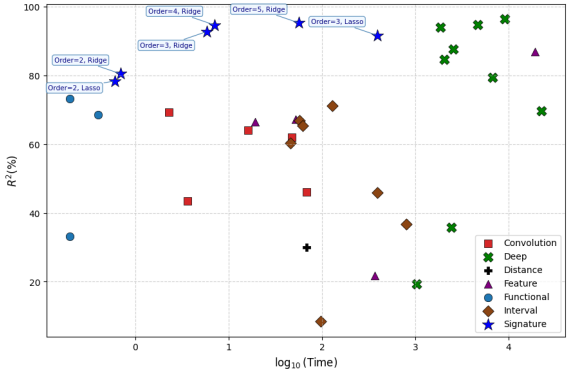






Empirical study of residuals. Results are obtained out-of-sample with the truncation order equal to 4.

Comparison with other ML models



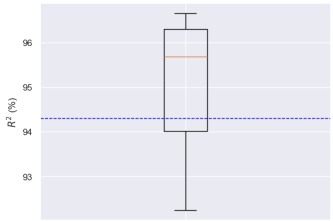
Benchmark. Comparison with methods from the *aeon* library [3].

Robustness with respect to the ESG distribution

- $\beta^* = \arg\min_\beta \mathbb{E}[|\Phi(X) - \langle \beta, \widehat{\mathbb{X}}_T^{\leq N} \rangle|^2] \rightsquigarrow$ sensitive to the distribution of X.

What happens if β^* is estimated under a given ESG distribution (e.g. calibrated to current market data), and the ESG distribution is subsequently perturbed without retraining the surrogate? (changes in market conditions or in the ESG specification) ?

Robustness with respect to the ESG distribution





Robustness under ESG parameter perturbations. The boxplots report the distribution of out-of-sample R^2 values and VIF relative errors across 50 perturbed ESG parameterizations. For each parameterization, 1000 new economic scenarios are generated and evaluated with the full ALM model. The signature-based surrogate is kept fixed after calibration on the central ESG distribution. The dotted horizontal line corresponds to the value obtained under the central ESG distribution on the reference dataset of 600 scenarios.

Takeaways

1. **Signatures provide a natural representation of path-dependent ALM outputs.**

   A complex nonlinear functional of an economic scenario can be approximated by a linear model on signature features.

2. **Low-order signatures already provide high predictive accuracy at very low computational cost.**

   In our experiment:

$$N = 4 \qquad \Longrightarrow \qquad R^2 = 94.63\%, \qquad \text{training time} = 7.1 \text{ s}, \qquad \text{VIF error} = -0.2\%.$$

3. **The surrogate can be reused under moderate shifts away from the ESG distribution on which it was trained.**

   It remains accurate under moderate ESG perturbations without retraining.

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Thanks for your attention!

Questions?